EMPHASIZING VISUAL RECOGNITION OF THE AMERICAN SIGN LANGUAGE ALPHABET

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Introduction



- **Vital Communication: Sign language is the primary voice for the Deaf community.**
- **Global Scale: WHO reports 1.5 billion people currently have hearing loss [8].**
- **The Barrier: 95% of deaf children are born to non-signing parents [9].**
- **Severe Impact: This gap causes high unemployment and educational disparities [10].**
- **The Solution: AI-based ASL detection is essential to bridge this divide.**

Problem Statement

1

A significant communication gap exists between the deaf community and the hearing majority, as most people do not understand sign language.

2

A computerized sign language recognition system is needed to bridge this gap, because relying on human translators at all times is impractical.

3

Creating this system is very difficult, as sign language is visually complex and has a large "inter-class variation," meaning many signs look very similar.

Objectives

1

Develop a high-accuracy classifier for 24 static ASL alphabets using Deep Learning.

2

To enhance the model's robustness using advanced data augmentation techniques, ensuring it performs well under various lighting conditions, angles, and backgrounds.

3

Deploy an interpretable Decision Support System (DSS) with visual feedback mechanisms.

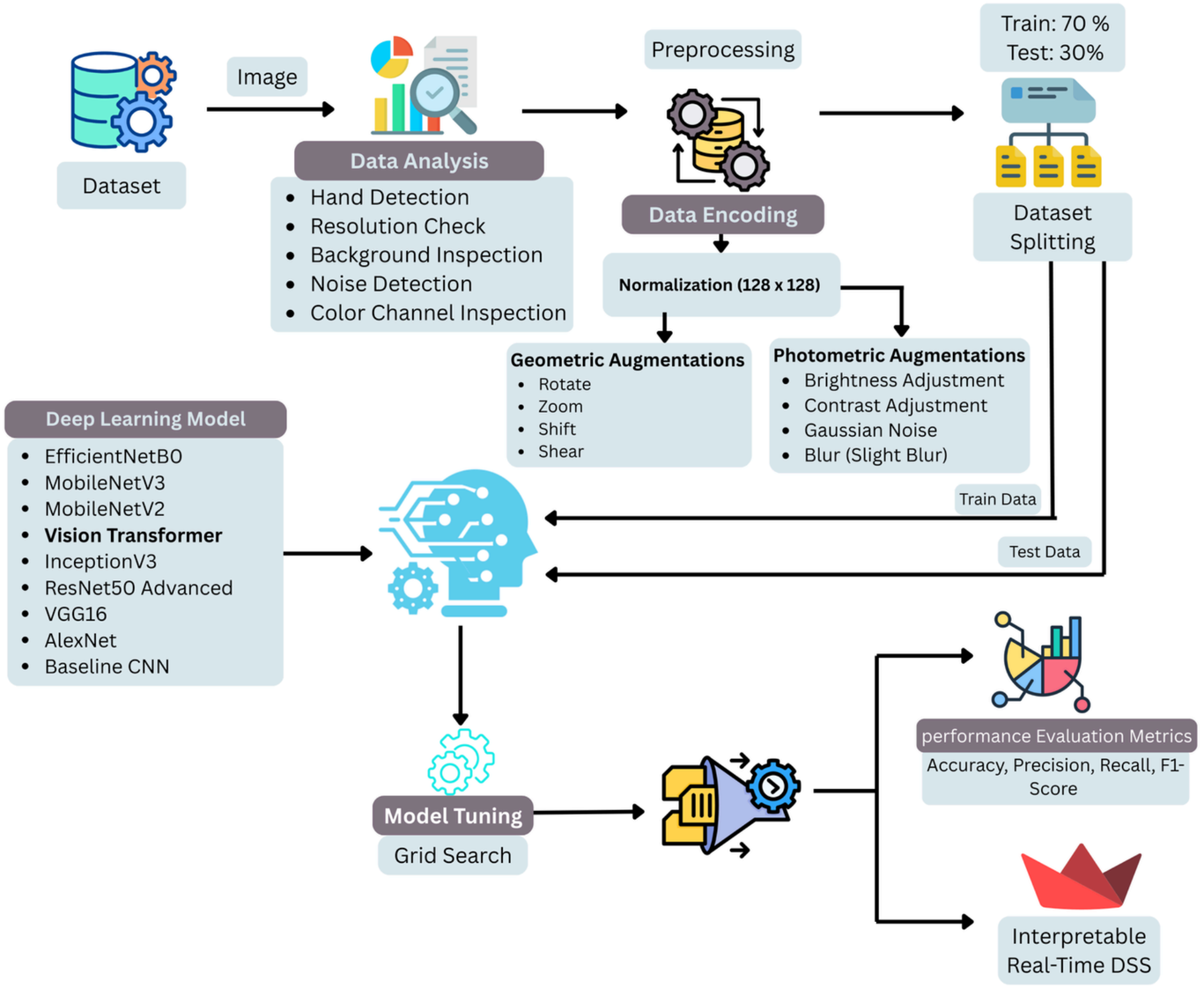
Related Work

Author(s) (Reference)	Methodology Used	DSS	Accuracy / Result
Sharmila Konwar et al. [1]	Used HSV color model to detect hand gesture color and edge detection technique for shape.	No	Achieved a 65% ASL recognition rate.
Vivek Bheda et al. [2]	Proposed a method based on a deep convolutional neural network (CNN) architecture.	No	Achieved 82.5% accuracy on alphabet gestures and 97% on digits.
Srinath et al. [3]	Presented a segmentation technique-based classification approach.	No	The model recognized ASL alphabets with a success rate of 86.67%.
Jayshree et al. [4]	Proposed a hand gesture recognition system based on the edge orientation histogram technique.	No	Achieved 88.26% accuracy for the recognition of ASL alphabets.
Joyeeta Singha et al. [5]	Used Eigenvalues and Euclidean distance for classification on a dataset of indigenous signs (ISL).	No	92.5% accuracy was achieved by this technique.
G. A. Rao et al. [6]	Proposed a Convolutional Neural Network (CNN) architecture for sign language recognition.	No	Achieved a 92.88% recognition rate.
Dhruv Rathi et al. [7]	Recommended a system based on the Transfer Learning approach, using MobileNet and Inception V3 models.	Yes	Achieved 95.06% and 93.36% accuracy, respectively.
Our Model	EfficientNetB0, MobileNetV3, Vision Transformer, InceptionV3, VGG16, Baseline CNN	YES	Achieved 94.10% with a real time support system

RESEARCH GAPS IN EXISTING STUDIES :

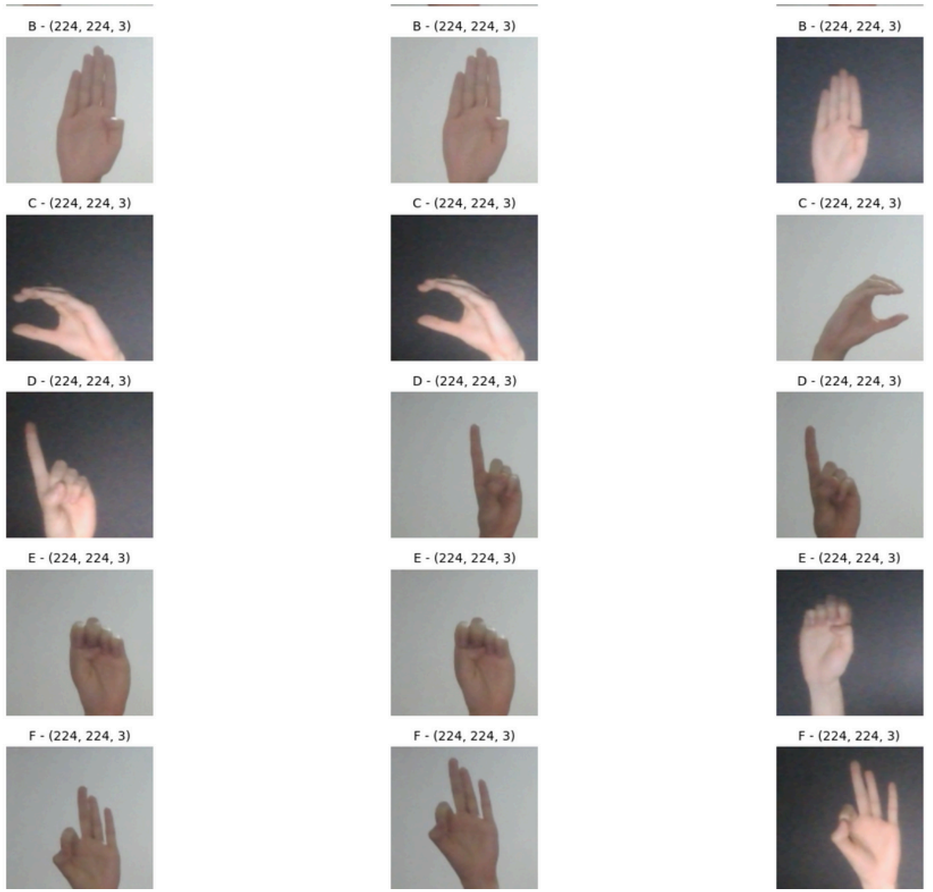
- **CNNs focus on local features**
- **Struggle to distinguish similar gestures (e.g., 'M', 'N', 'S').**
- **Lack of real-time Decision Support Systems (DSS)**
- **Limited dataset**

Proposed Methodology



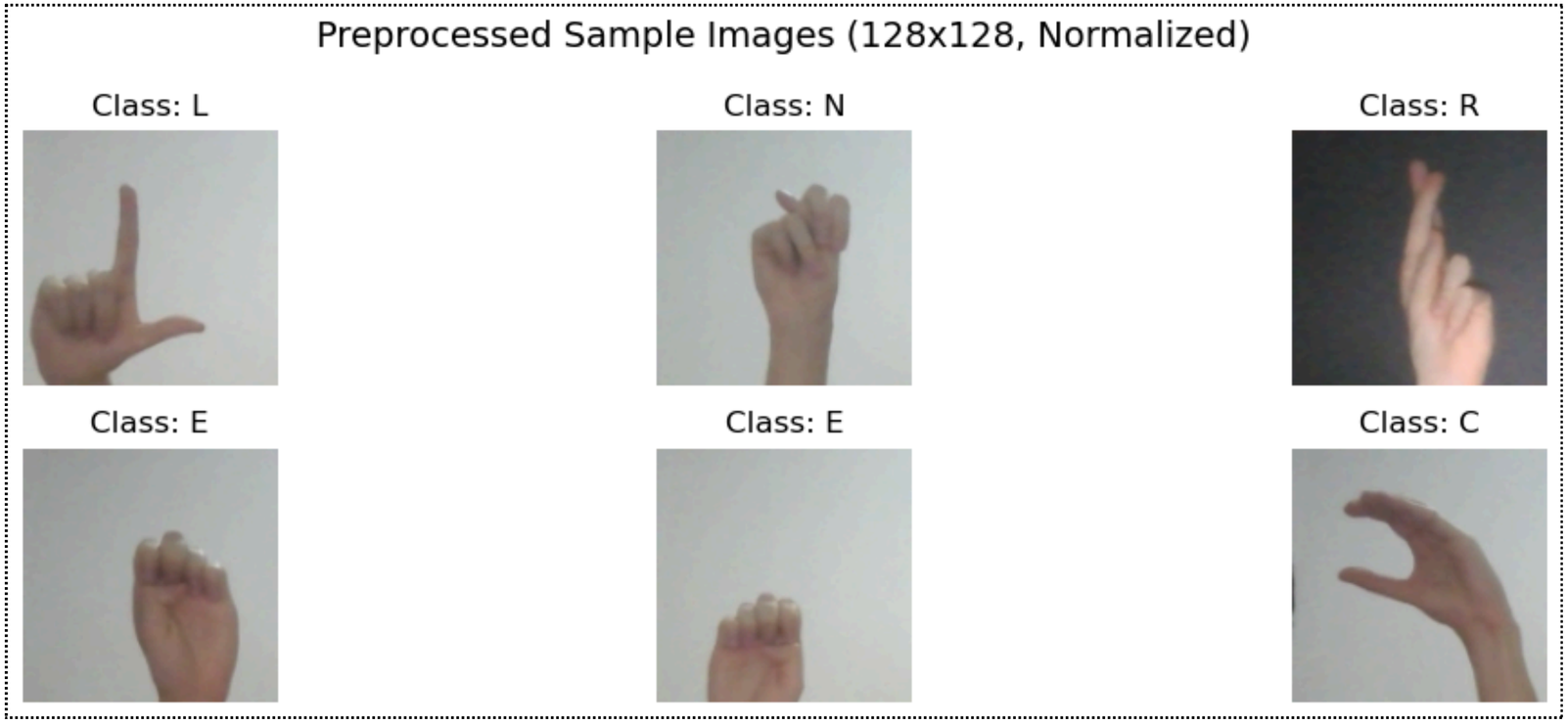
Data Description

- **Source:** Kaggle ASL Alphabet
- **Size:** 2427 images
- **Average Images per class:** 101
- **Format:** RGB color images (JPG).
- **Resolution:** 224 × 224 pixels



Core Preprocessing (1/4)

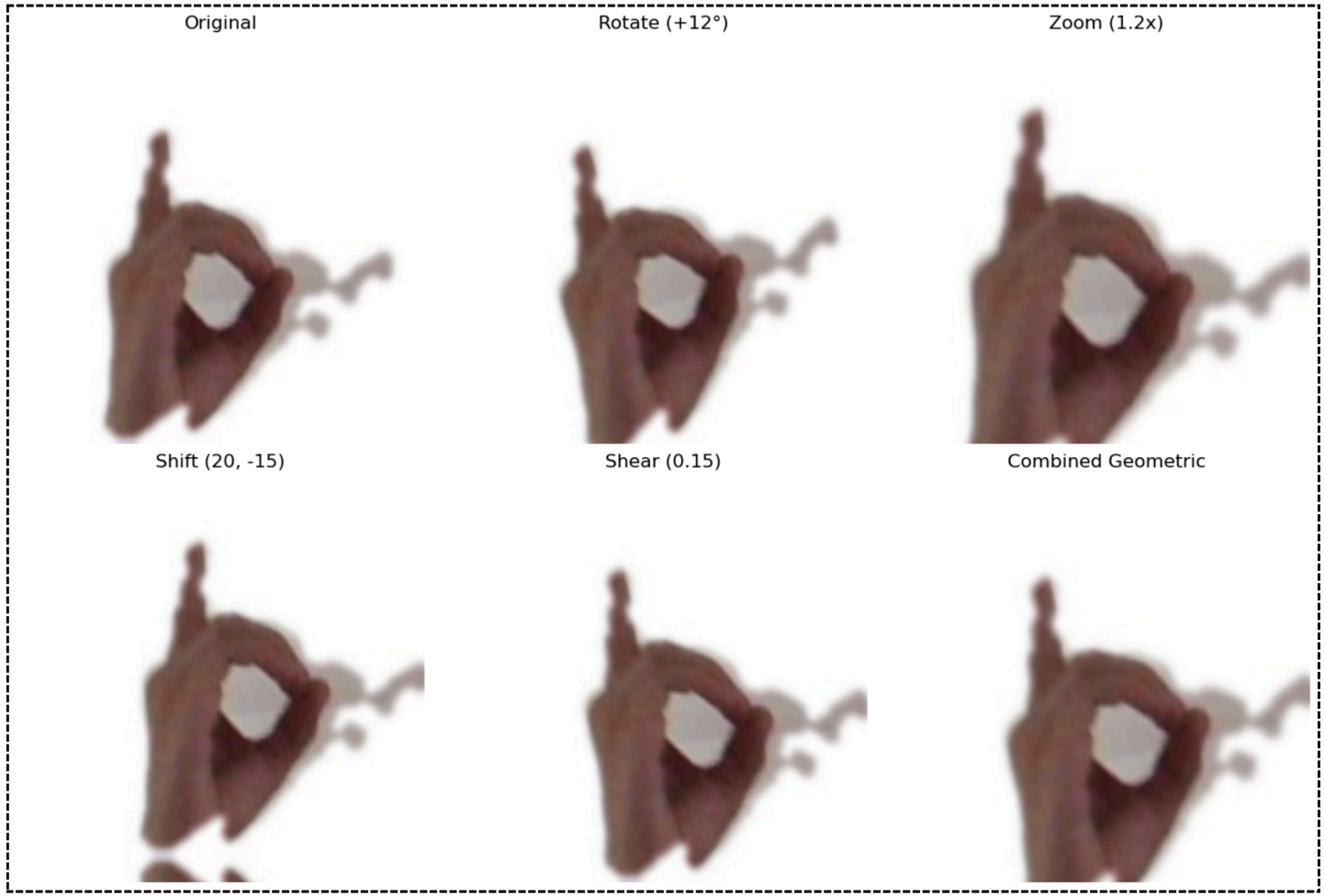
- **Loading**
- **Resizing**



Core Preprocessing (2/4)

Geometric Data Augmentation

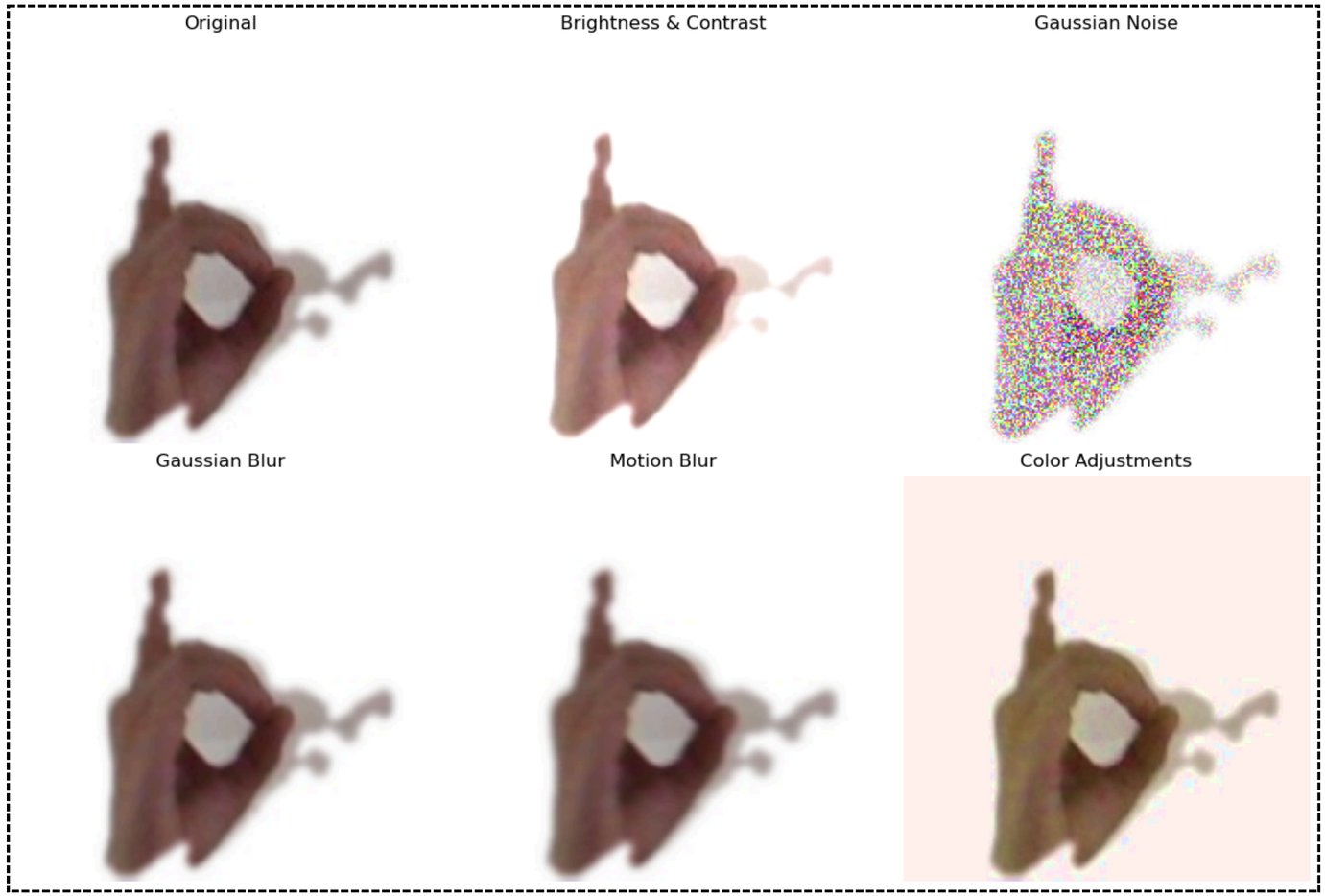
- **Rotation (–12° to +12°)**
- **Zoom (4x)**
- **Width & Height Shift**
- **Shear Transform**



Core Preprocessing (3/4)

Photometric Data Augmentation

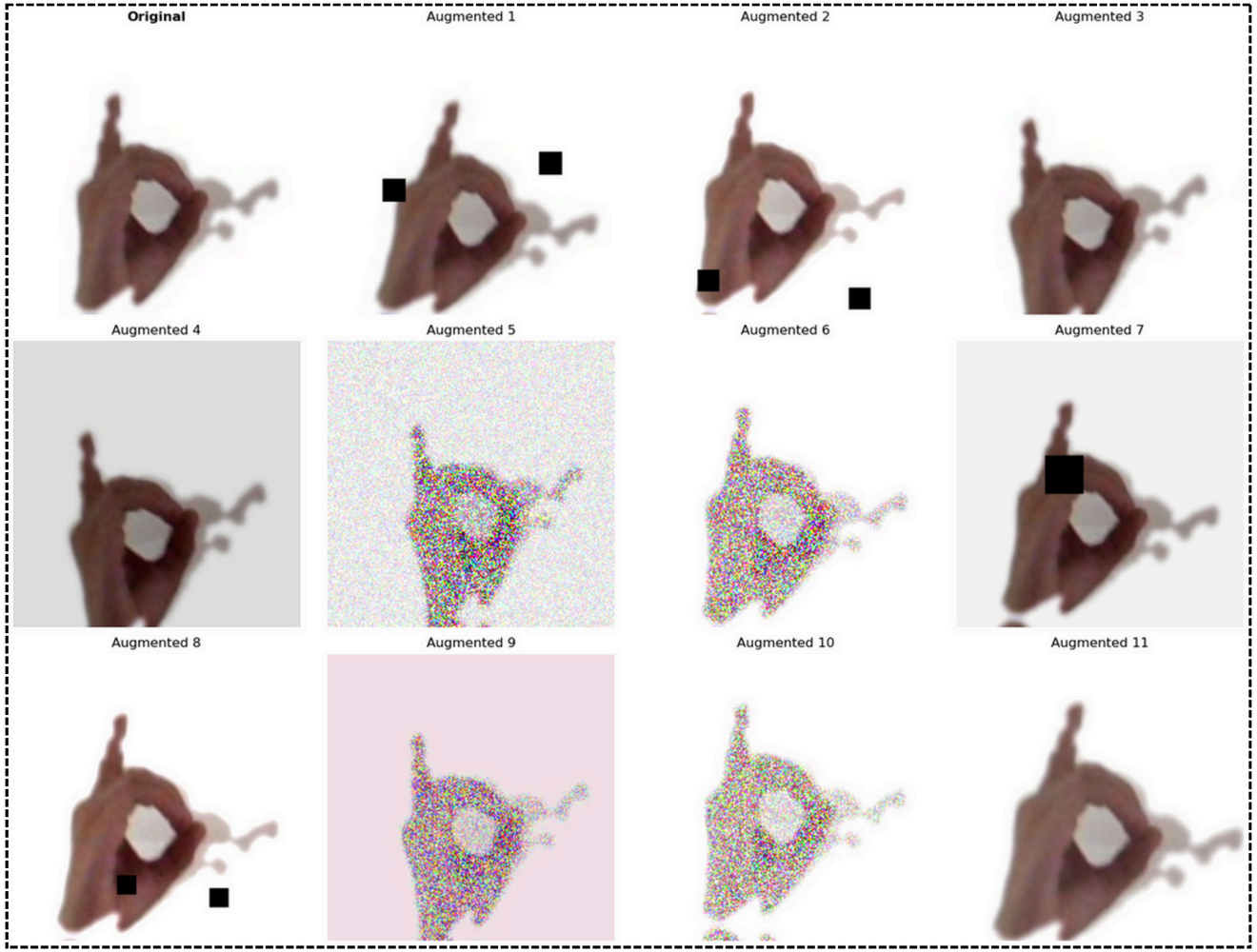
- **Brightness & Contrast Adjustment**
- **Gaussian Noise**
- **Gaussian Blur**
- **Motion Blur**
- **Color Adjustments**



Core Preprocessing (4/4)

Data Augmentation on ASL Hand Images

- Original
- Augmented 1, 2, 8 (black patches)
- Augmented 3, 4 , 11 (Color Adjustments)
- Augmented 5, 6, 9, 10 (Gaussian noise)
- Augmented 7 (Random Occlusion)

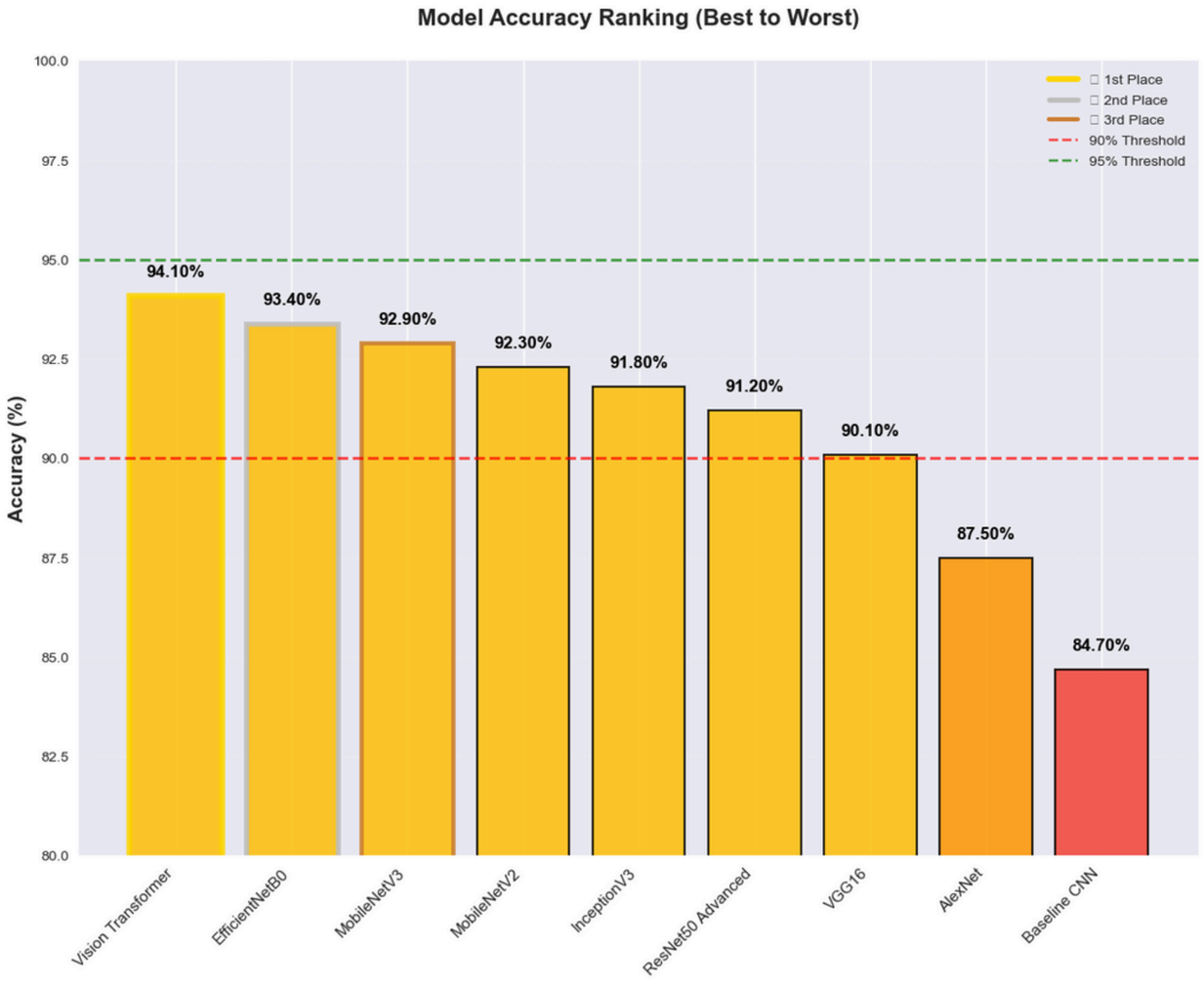


Optimized Hyperparameters

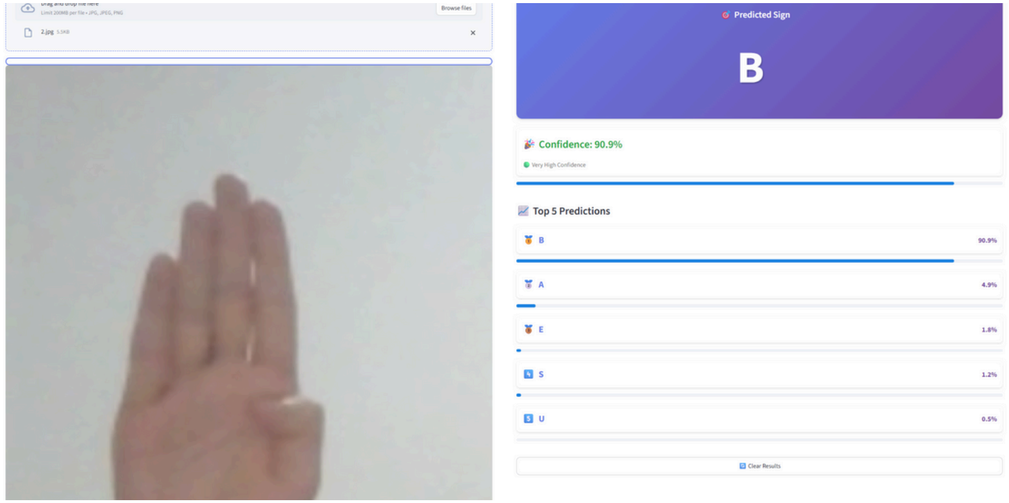
Model	Optimizer	Learning Rate (Initial)	Fine-tuning LR	Epochs (Initial)	Final Tune Epochs	Batch Size	Dropout Rate
Baseline CNN	Adam	0.001	N/A	15	N/A	32	0.5, 0.3
MobileNetV2	Adam	0.001	0.0001	10	10	32	0.3, 0.2
MobileNetV3	Adam	0.001	0.0001	15	10	32	0.4, 0.2
InceptionV3	Adam	0.001	0.0001	15	10	32	0.5, 0.25
Vision Transformer (ViT)	Adam	0.0005	N/A	10	N/A	32	0.5, 0.1
AlexNet	Adam	0.001	N/A	15	N/A	32	0.5
EfficientNetB0	Adam	0.001	0.0001	15	10	32	0.5, 0.25
VGG16	Adam	0.001	0.0001	15	10	32	0.6, 0.3
ResNet50	Adam	0.001	0.0001	10	N/A	32	0.5, 0.3, 0.2

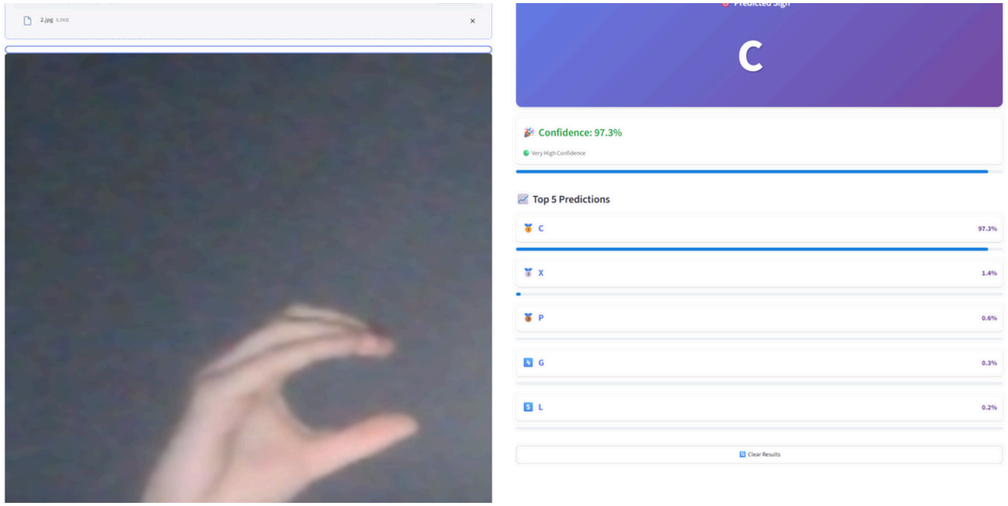
Results

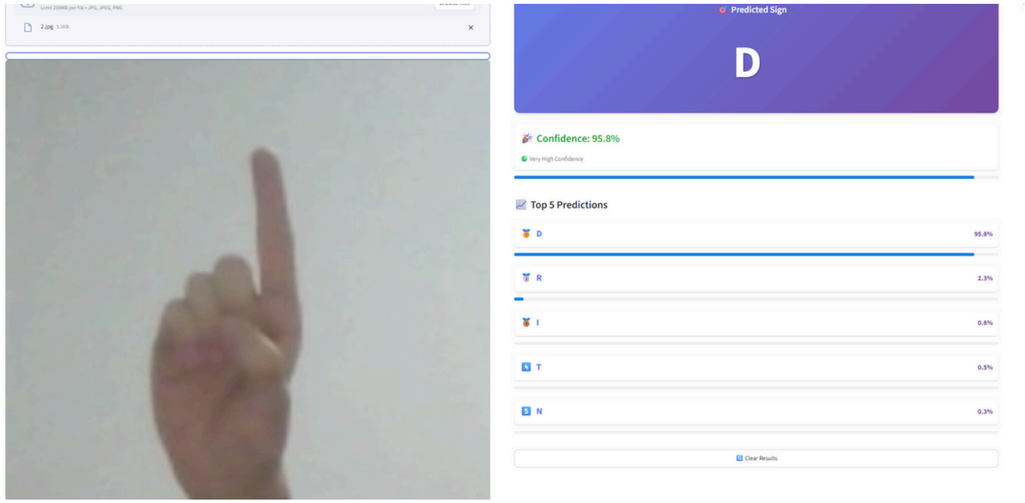
Model	Accuracy	Recall	F1-Score	Precision
Vision Transformer	0.941	0.941	0.945	0.95
EfficientNetB0	0.934	0.934	0.938	0.943
MobileNetV3	0.929	0.929	0.933	0.938
MobileNetV2	0.923	0.923	0.927	0.932
InceptionV3	0.918	0.918	0.922	0.927
ResNet50	0.912	0.912	0.916	0.921
VGG16	0.901	0.901	0.905	0.91
AlexNet	0.876	0.876	0.88	0.885
Baseline CNN	0.847	0.847	0.851	0.855

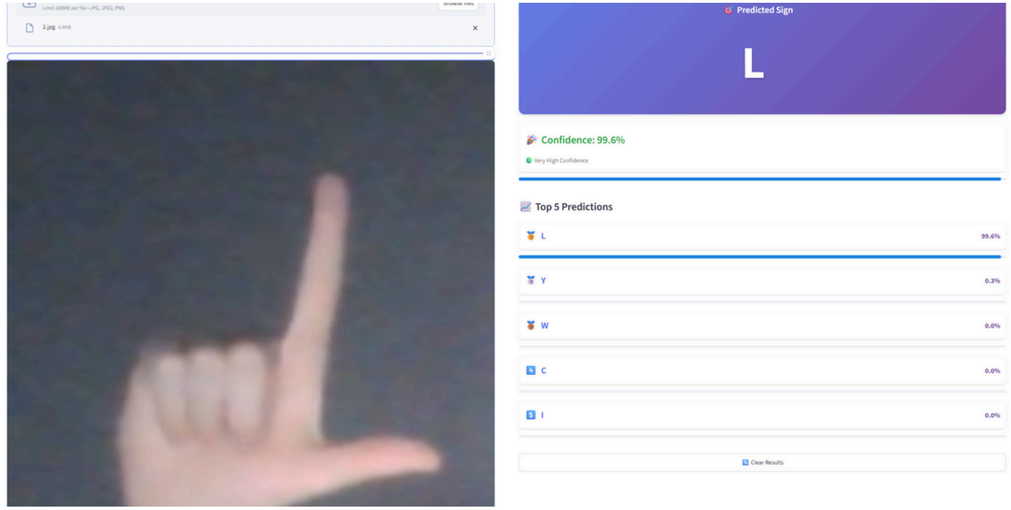


Real Time DSS









Conclusion

- The Vision Transformer (ViT) model was the most successful, achieving the highest performance with 94.10% accuracy, demonstrating its superior ability to interpret complex ASL features.

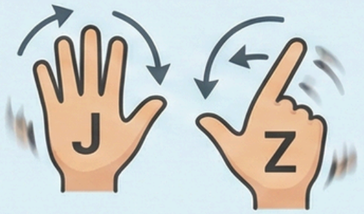
- Advanced models like EfficientNetB0 and MobileNetV3 were also highly effective, providing a strong balance of high accuracy and potential for efficient use.

- The project successfully created a foundation for a practical tool that can detect ASL alphabet letters.



FUTURE WORK

DYNAMIC GESTURE INTEGRATION

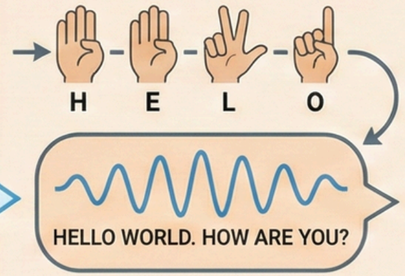


Incorporate dynamic gestures (e.g., J and Z) to complete full alphabet recognition.

Current: 24 Static

Future: Complete Alphabet

CONTINUOUS RECOGNITION



Extend from isolated letter detection to word- and sentence-level recognition.

MOBILE DEPLOYMENT



Develop a mobile application for real-world usability.

BDSL ADAPTATION



Adapt the system for Bangla Sign Language (BdSL) support.

Future Impact

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THANK YOU